

Guidelines for Bachelor/Master theses and Applied Research Projects (ARP)

For any type of scientific work under the supervision of
Prof. Dr.-Ing. Stéphane Danjou

Updated: 11.08.2023

Content

1	Introduction	1
2	Basics for Bachelor/Master Theses and ARPs.....	3
2.1	Differences Between Bachelor Theses, Master Theses and ARPs.....	3
2.2	Steps of Writing Bachelor/Master Theses	4
3	Guidelines to Prepare Theses and ARPs	6
3.1	General Guidelines Regarding the Form and Content	6
3.2	Structure	6
3.2.1	Introduction	7
3.2.2	Literature review	8
3.2.3	Methodology	8
3.2.4	Results	10
3.2.5	Discussion	10
3.2.6	Conclusion.....	11
3.2.7	Summary and Outlook.....	11
3.2.8	Bibliography or Reference List	11
3.2.9	Appendix.....	11
3.3	General Considerations	11
3.3.1	Technical terms	11
3.3.2	Abbreviations	12

3.3.3	Objective vs. Subjective Language.....	12
3.3.4	Choosing a 'Voice'	12
3.3.5	Personal vs. Impersonal.....	13
3.3.6	Verb tenses in academic writing – by section	14
3.3.7	Sentence length	18
3.4	Citation Method	19
3.4.1	References in the text	20
3.4.2	Online references in the text.....	21
3.4.3	List of references.....	21
3.5	Acknowledging, Citing and Referencing Use of AI Tools and Technologies.....	22
3.6	Further Comments in the Text.....	23
3.7	Figures and Tables.....	24
3.8	Mathematical Equations	25
3.8.1	Formatting Mathematical Equations.....	25
3.8.2	Numbering.....	26
3.8.3	Units.....	27
3.9	Guidelines Regarding the Format.....	27
4	Common Mistakes in Theses and ARPs	29
4.1	Formal Mistakes	29
4.2	Word Processing Hints	30
5	Evaluation	31
5.1	Evaluation Criteria for Theses.....	31
5.2	Evaluation Criteria for ARP	31
5.3	Evaluation of the Colloquium	32

1 Introduction

Writing is a very important part of science; it is used to document and communicate ideas, activities and findings to others. Scientific writing can take many forms from a lab notebook to a project report, or from a paper in an academic journal to an article in a scientific magazine.

The following notes are intended to help during the writing of scientific papers at the HSRW. In addition to these notes there will be an intensive support by the advisor¹. You should use this opportunity to discuss your work progress regularly. Furthermore, the advisor also supports the creation of the structure of your paper and provides feedback regarding its content. Furthermore, the advisor will support you with suggestions regarding the creation and conduction of presentations.

This guide starts with basic information, differentiated between bachelor/master theses and ARPs. Afterwards, there is no differentiation since the information is necessary for both, bachelor/master theses and ARPs. This part starts with formal guidelines for the preparation of scientific work and subsequently reveals common errors that should be avoided.

Good scientific writing is:

- **clear** - it avoids unnecessary detail;
- **simple** - it uses direct language, avoiding vague or complicated sentences. Technical terms and jargon are used only when they are necessary for accuracy;
- **impartial** - it avoids making assumptions (Everyone knows that ...) and unproven statements (It can never be proved that ...). It presents how and where data were collected and supports its conclusions with evidence;
- **structured logically** - ideas and processes are expressed in a logical order. The text is divided into sections with clear headings;

¹ For reasons of readability only the male form is used in this document although these statutes address all members - the female form is explicitly included.

- **accurate** - it avoids vague and ambiguous language such as about, approximately, almost;
- **objective** - statements and ideas are supported by appropriate evidence that demonstrates how conclusions have been drawn as well as acknowledging the work of others.

To reflect the characteristics of good scientific writing in your own work, you need to think about the way that you write and the language that you use. A good scientific author will have given consideration to the following choices in writing, making decisions that improve the effectiveness of the writing.

2 Basics for Bachelor/Master Theses and ARPs

2.1 Differences Between Bachelor Theses, Master Theses and ARPs

Theses (bachelor and master) differ in terms of editing time, length and the level of complexity and/or scientific demand. Table 2-1 shows an overview of these differences.

Table 2-1: Differences in editing final papers

	Editing time	Length ²	Presentation of the thesis/ARP	Second examiner
Bachelor thesis	3 months	50 - 70 pages	20 minutes presentation of the thesis	Second examiner is a company supervisor if the thesis is done in cooperation with a company. Otherwise a second examiner has to be arranged.
ARPs	4 months	about 40 pages	No presentation	No second examiner
Master thesis	4 months	60 - 80 pages	20 minutes presentation of the thesis	Second examiner is a company supervisor if the thesis is done in cooperation with a company. Otherwise a second examiner has to be arranged.

In addition to these formal aspects, different types of final papers can be distinguished:

- Theoretical papers
 - Literary papers
 - Papers which focus on the development of models
- Practical work
 - Empirical work including survey and analysis

² The length of theses can be seen as a rough benchmark and refers to the pure content without lists (content, list of figures, list of tables, list of abbreviations, and list of references) and appendices. For example, if a bachelor thesis is completed after 35 pages of content, it should not be extended to 40 pages just to meet this number. A master thesis can also include more than the specified pages on account of relevant content. If you can describe the issue precisely in fewer pages, you should do this.

- Work including (partial) implementation of prototypes
- Work with practice partners

Topics for practical work and contacts for practice partners are maintained by the professor. Own topic suggestions for practical work topics will only be accepted in exceptional cases and requires that a professor or research associate will mentor that topic and the professor agrees with it.

2.2 Steps of Writing Bachelor/Master Theses

- If you want to write a thesis at a professor, please contact him early enough. Each professor/research associate is responsible for specific topics.
- Topics for bachelor/master theses are regularly published. Students can apply to the specific advisor for a topic.
- To write a bachelor/master thesis, the thesis needs to be officially registered via SharePoint (<https://sharepoint.hochschule-rhein-waal.de/sites/Ab-schlussarbeit/SitePages/Home.aspx>). Please agree on the topic and further details with your supervising professor before registration.
- The starting point for the bachelor/master thesis can be adjusted with the advisor. Thus, it is also possible to use the lecture-free time for writing bachelor/master thesis.
- After the reception of the topic, orientation, and literature collection, you should consult your advisor to clarify whether your interpretation of the topic is correct or not.
- After this first orientation phase, you should discuss the structure with your advisor. The structure should include information regarding the length of each section in order to identify wrong focus areas.
- After the discussion with the advisor there is a final discussion with the professor about your structure (approx. two weeks after the official start of processing).
- During the preparation of the bachelor/master thesis you should maintain regular consultations with your advisor. Your structure can change during the time, but the agreed objective should stay.
- There is a further appointment with the professor approx. three weeks before the deadline for submission. Topics in this appointment are the current progress, important results and problems.

- Before contacting other parties (e.g. industrial establishment, agencies, or other institutes) it is necessary to consult your advisor.
- During the real editing phase, you need to calculate with enough time for writing and editing the bachelor/master thesis. You should start as early as possible with writing the text. It is also helpful to discuss finished parts with your advisor.
- If you recognize that you can't finish your work based on time-dependent problems, please contact your advisor early on.
- According to the examination regulations, a presentation of your results is a part of the bachelor/master thesis. For the conditions of the presentation you need to contact your advisor.
- The delivery of the bachelor/master thesis is done in two bounded hardcopies at the examination office and one softcopy submitted via SharePoint. The hardcopies will be transferred to the supervisors.
- The announcement of your grade takes place after the colloquium. You will get feedback regarding the criteria, which were vital for the assessment. Afterwards, your grade will be delivered to the examination office.

3 Guidelines to Prepare Theses and ARPs

In this chapter, guidelines and advice for bachelor/master theses and ARPs are given. The contents are valid for both types of papers; exceptions from this are mentioned explicitly.

3.1 General Guidelines Regarding the Form and Content

The following guidelines apply for the content and formal design of the written part:

- Concentrate on the main subject and therefore refrain from long observations of general nature. For example, it does not make sense to write long introductions about the general use of data processing in the industrial sector, if the topic is about a special problem of data processing in material management within an industrial organization.
- Write for a professional reader and not for layman. This reader has the general knowledge about the problem and is interested in a presentation of the main results. Thus, the scientific paper should only include definitions of the core content. For example: In a thesis about technical design calculations a general definition or introduction to the fundamentals of mechanics is not necessary.
- A compact representation and good illustration of content can be reached with diagrams, tables and listings (particularly pros and cons).
- It is not good practice to use platitudes (“Based on the increasing competition, rationalization becomes more important.” or “An entrepreneur who wants to stay successful, must work with costs as low as possible”).

3.2 Structure

The structure of the text should be justified as factual as possible and well structured. There should be only subsections if there are at least two subsections. You should also remember that a consistent level of all subsections is necessary, i.e., use second or third level continuously. The subdivisions in the main part should reflect the methodological approach. In addition to that, the subsection headline should match the content and form of the upper headline, but it may not be the same. Furthermore, the topic of the work may not be equal to a subsection headline.

The table of content should be numeric. There have to be a dot between each number, but the last number is not followed by a dot. Example: 3.2.1

You should use breaks to distinguish different aspects in one chapter.

The table of content should entail:

- List of figures and maybe a list of tables
- List of abbreviations
- Structure of the main body
- Bibliography / List of references
- Appendix (optional)

The numeration of the lists before the main text as well as the list of the references is done with continuously Roman numerals. The numeration starts with (“I”) on the cover sheet (Important: The page number should not be printed on the cover sheet of the final paper. So use suitable settings in your processing program or use this document as a template.). The pages of the text and the enclosure should be Arabic and start with “1”.

In the following you will get a brief overview of the different sections of the main body and the ending part of your report.

3.2.1 Introduction

The introduction chapter of your report outlines its core arguments, hypotheses, and expected results. It contains adequate background information on your topic of interest. Furthermore, it establishes the relevance of your thesis by highlighting its contribution to the knowledge base of its topic. Writing a gripping introduction helps the readers understand the context of your thesis. The introduction of a thesis/ARP should have – depending on the topic – all or some of the following stages:

- State the general topic and give some background
- Define the terms and scope of the thesis topic, if applicable
- Outline the existing situation
- Evaluate the current situation and identify the gaps
- Identify the importance of the proposed research
- State the main research questions

- State the purpose of the study and/or research objectives
- Clearly set the boundaries of your research work

3.2.2 Literature review

The literature review chapter sets the premise of your scientific paper. It examines and evaluates the research works that's been conducted so far on your topic and passively highlights the contributions of your work.

A literature review is a survey of academic sources on a specific subject, providing an overview of current knowledge, allowing you to discuss relevant theories, methods, and gaps in the existing research. Writing a literature review contains finding relevant publications, critically analyzing the sources, and explaining your findings in the literature. A well-written literature review doesn't only summarize sources, it also aims to analyze, synthesize, and critically evaluate to give a clear picture of the state of knowledge on the topic.

To write an impeccable literature review, consult a plethora of sources and mention the canon related to your topic. Also, put forward your review in a logical, chronological, and structured manner to better outline the knowledge gaps in your field of study and how your own work will fill them.

The following simple and straightforward tips can act as the exhaustive rubric and offer meaningful insight on how to formulate a flawless literature review:

Step 1. Probe similar works for a well-structured literature review

Step 2. Analyze, not just synthesize: provide a detailed critique of the subject

Step 3. Organize your literature review systematically

Step 4. Establish the purview: specify the scope of the literature review

Step 5. Abstain from plagiarism

3.2.3 Methodology

As the name suggests, the methodology section of a scientific paper consists of all methods and procedures you have used in your thesis/ARP. A well-written methodology accentuates

the plausibility of your research methods. In addition, it enables readers to understand why you chose specific methods and how they are justified for your research.

To garner more credibility, you can include the pitfalls and difficulties associated with your choice of research methods. The methodology section is an unavoidable part of a thesis or a research paper. Considering errors in the methodology section enervates the entire thesis.

Follow the steps below to write a good methodology section for a thesis/ARP:

- (1) Give an outline of the research design
- (2) Don't forget to define the philosophy behind the research
- (3) Mention the research approach
- (4) Introduce the research methods
- (5) Note the following points to highlight in the methodology. No matter what methodology you have chosen, you have to focus on the following points:
 - Explain sampling strategy.
 - Clearly state the procedure of the research paper.
 - Mention how you collect the data. (Data collection)
 - Explain how data are analyzed for your research. (Data analysis). Suppose you have written in qualitative strategy like thematic analysis, mention the researcher you have followed.
 - Mention the validity of the data and result.
 - Discuss all ethical aspects of your research paper.
- (6) Most important tips to compose an impactful methodology
 - Don't drift from your objective and the purpose of your research.
 - Explore research papers and their methodology sections to have a better idea.
 - Make sure to properly cite relevant sources. Check the section about referencing and citation methods.
 - Refer to all the hurdles you have experienced while writing your report.
 - Make sure to rectify grammatical and punctuation errors.
 - Ensure that the section is readable and doesn't consist of long and complex sentences. Long sentences can hamper the tone of the methodology.

3.2.4 Results

This section comprises the outcomes of your research work. It includes all the observations you made and the answers to all your hypotheses in the thesis/ARP. When writing the “results” chapter, include only factual data and format it to be distinguishable. Use tables, graphs, subheadings, and generic comments for the results. The aim is to enable your readers to discern the result of your research.

3.2.5 Discussion

The discussion chapter of your report should begin with a brief summarization of the outcome of your research work. It should explain how your results address your hypotheses and highlight any repetitions in your observations. You can also add comments on how you want the readers to interpret your results and about your agreements and disagreements with the available research work in your field.

Writing a flawless thesis/ARP requires much more than only subject matter expertise. It requires expertise, experience, and in-depth thinking, along with sharp intelligence. Though most students add a discussion chapter in their report, many of them end up messing up the essay or missing out on the central issues.

A discussion chapter in a thesis/ARP is a place where you have the chance to delving into the analysis, importance, and relevance of your research. This section focuses on explaining and analyzing what you have researched, presenting how it is associated with the existing literature. It is also a place for arguments supporting your entire discussion.

The following tips can help you design your discussion chapter:

- Understand the objective of your thesis
- Refer to hypotheses and literature review
- Evaluate your results and compare them with existing studies
- Understand the limitation of your research

The following 5 questions might be helpful to write a sound discussion section:

- How well do you understand the objective of your study?
- What message is conveyed by your results?
- How do your findings compare to findings in literature?

- Why should your findings matter?
- In what light should your findings be viewed?

3.2.6 Conclusion

The final section of your document consists of a precise answer to your hypothesis. In addition, the “conclusion” chapter of your thesis/ARP should stress the achievement of the aims of your work. You should also include certain limitations of your research to convey the fact that there is still scope for further research in your field.

3.2.7 Summary and Outlook

Briefly summarize your work and give some prospective steps. Could your work be extended or continued in further future theses or ARPs? Is there anything that could be done after your thesis/ARP to get better results?

3.2.8 Bibliography or Reference List

3.2.9 Appendix

In this chapter you can add any kind of material that is just supplemental and not explicitly necessary for the understanding of the work. If needed, it is possible to create several appendices, e.g. naming Appendix A, Appendix B, Appendix C, etc.

3.3 General Considerations

To make your writing clear, accurate and concise you should consider carefully the words that you use, and the ways in which you use them.

3.3.1 Technical terms

In most scientific writing you will need to use some scientific or technical terms in order to be clear and unambiguous. However, use such terms only when you need to do so and do not try to impress the reader by using unnecessary technical jargon or lengthy words.

3.3.2 Abbreviations

Abbreviations can be a very useful way of saving time and avoiding repetition, but they can be confusing and might not be understood by everyone. Use standard abbreviations where these exist, and reduce your use of abbreviations to an absolute minimum; they are rarely essential. Write the term which you intend to abbreviate in full when it occurs the first time. From this point on you can use the abbreviation instead.

As an example:

Additive Manufacturing (AM) is an appropriate name to describe the technologies that build 3D objects by adding layer-upon-layer. AM is getting faster and more reliably and growing into a mainstream manufacturing technique not just for prototypes.

3.3.3 Objective vs. Subjective Language

Objective language is language that is impartial and states a fact or process; subjective language is open to question or interpretation as it implies personal thought or belief. For example:

objective *The car travelled at 38 kilometres per hour*

is a clear, objective statement of fact. However:

subjective *The contents of the test tube turned a beautiful blue colour*

uses *beautiful* in a way that is subjective because it cannot be measured or accurately explained to the reader. Always use language that is concrete and specific rather than vague and personal.

3.3.4 Choosing a 'Voice'

Scientific writers have a tendency to use *passive* rather than *active* expressions; stating that *a* was affected by *b* uses the passive voice while stating that *b* did something to *a* uses the active voice. The following example shows a sentence written in both the passive and active voices.

passive *The experiment was designed by the research officer*

active *The research officer designed the experiment*

The passive voice is particularly useful when:

- you wish your writing to be formal and depersonalised:

passive *It was agreed that the experiment should be...*

active *We agreed that the experiment should be...*

- information about the agent is obvious or unimportant:

passive *Extra solvent was added to the flask*

active *The technician added extra solvent to the flask;*

- you do not know the identity of the agent:

passive *The water pipe was broken in three places*

active *Something/someone had broken the water pipe in three places*

However, the use of the passive voice can lead to clumsy and overcomplicated sentences.

passive *Difficulty was experienced in obtaining the product in a high state of purity*

is rather convoluted way of saying

active *The product was difficult to purify*

which is a much clearer and more straightforward statement.

In general, the active voice is clearer, more direct and easier to read, but the passive voice can be more appropriate in particular circumstances. What is most important is for you to be aware of how you are writing, and how the voice that you choose affects the tone and the meaning of your words.

3.3.5 Personal vs. Impersonal

Scientific writers often try to avoid the use of personal expressions or statements in order to make their writing seem more impartial and formal. The following sentence has been written with both personal and impersonal expressions to highlight the contrast between the two writing styles.

impersonal *The explanation for this phenomenon may be found in...*

personal *We/I believe that the explanation for this phenomenon may be found in...*

However, used indiscriminately, writing impersonally can result in clumsy statements through an excessive use of the passive voice. This can lead to ambiguity or inaccuracy in your written work, for example:

impersonal & passive *It was decided that the temperature should be raised*

gives no information about the identity of the people who made the decision.

personal & active *We decided that the temperature should be raised*

avoids ambiguity and makes the sentence sound more direct, but uses the personal and rather informal we.

impersonal & active *The research team decided that the temperature should be raised*

is clear and direct.

Think carefully about your use of impersonal and personal expressions, taking care to ensure that your writing is always clear and unambiguous.

3.3.6 Verb tenses in academic writing – by section

Scientific writing frequently uses the past tense, particularly when the main focus of the writing is to describe experiments or observations that took place prior to the time of writing. However, the past tense may not be appropriate for everything that you write and sometimes you will need to combine different tenses in the same piece of writing. For example, the use of different tenses can help to clarify what happened or what you did in the past (past tense), what you conclude (present tense) and what will be an issue for the future (future tense). The following sentences show how different tenses can be used to achieve clarity in your written work.

The experiment was carried out in a sterile environment (past tense for a statement of what happened).

It is particularly important to avoid contamination (present tense for a statement that is a general 'truth').

It will be necessary to ensure that the same conditions are replicated in future experiments (future tense for a recommendation for the future).

An appropriate use of present simple, past simple, and present perfect tenses can contribute to a clear and unambiguous writing style. Here you can find general guidelines (taken from <https://proofreading.org>) on which verb tenses to use in each section of a thesis.

ABSTRACT OR SUMMARY

Use the **present simple tense** to state what the paper investigates and what its goals are, and to report general facts that continue to be true.

Examples:

- This study *explores* the potential clinical uses of topical compounds.
- Membership in the European Union (EU) greatly *impacts* each member country's economic policies.

Use the **present perfect tense** to discuss past research that is relevant to the current study.

Examples:

- Past research *has shown* that the presence of local fisheries has a direct impact on wild salmon populations.
- Philosophers *have interpreted* this thought experiment differently.

INTRODUCTION

Use the **present simple tense** to explain what the paper is about, how it is structured, and why it's important, and to define terms and share facts that are widely known and won't change over time.

Examples:

- Chapter 4 *describes* the research methodology.
- The theory that all actions *are* intrinsically right or wrong *is known* as moral absolutism.

Use the **past simple tense** to provide historical background and context.

Examples:

- Rosalind Franklin *contributed* greatly to the study of the molecular structure of DNA.
- The literary and philosophical movement known as transcendentalism *began* in the late 1820s.

Use the **present perfect tense** to present past research that is relevant to your study.

Example: Previous studies *have analysed* the drug's effects on adults, but none has focused on adolescents.

Use the **present simple tense** to describe theories and define terms.

Examples:

- In anaerobic respiration, glucose *breaks down* without oxygen.
- A new theory *suggests* that the Big Bang singularity never occurred and that the universe may have had no beginning.

LITERATURE REVIEW

Use the **past simple tense** to discuss the findings of earlier studies.

Example: Stokes and Chesterfield *found* that doctors' treatment recommendations are often ignored by patients.

Use the **present simple tense** to state facts that do not change over time, to express opinions or propose a thesis statement, and to narrate the plot of a literary work.

Examples:

- Chronic illnesses *vary* in type and intensity.
- At the start of the novel, Ammu *moves* to Calcutta to live with a distant aunt.

Use the **present perfect tense** to introduce past research that is relevant to the current study, to address a gap in the current knowledge base, and to correct a dated theory that is no longer accurate.

Example: Previous studies *have found* that almost one quarter of teenagers experience cyberbullying.

METHODS AND RESULTS

Use the **past simple tense** to detail your experiments and to discuss events that began and ended in the past. If your paper must avoid using the first person, you will likely use the passive voice in this section, foregrounding the actions taken rather than the researcher(s) who took them.

Examples:

- Over 90% of samples *tested* positive.
- Participants *were divided* by age group.

Use the **present simple tense** to describe methodologies that don't change over time and to refer to figures, charts, tables, or other sections of the paper.

Examples:

- A geographic information system (GIS) *presents* all types of geographical data.
- Chart 3 *illustrates* the rapid growth of the portfolio's value.

DISCUSSION AND CONCLUSIONS

Use the **present simple tense** to interpret data or express an opinion, and to report facts that are unlikely to change in the future.

Examples:

- The results *support* the hypothesis.
- These results *imply* a strong connection between the variables.
- The company *does not plan* to develop artificial intelligence software.

Use the **past simple tense** to summarise the results of your own research or discuss studies conducted by others.

The study *found* that smoking adversely affects pulmonary function.

Limitations:

Use the **past simple tense** to explain the steps taken to conduct the study and its shortcomings.

Examples:

- Qualitative data *were collected* via surveys.
- The sample size *was* not large enough to be considered representative of the general population.

Use the **present simple tense** to state widely accepted facts that do not change over time.

Example: Self-reported information *is* of limited use because it cannot be verified independently.

Recommendations and implications:

This section discusses events that may happen in the future. However, no writer can make predictions with absolute certainty. To avoid or 'soften' the **future tense**, use modal verbs, such as *can*, *should*, and *may*; verbs that suggest uncertainty, such as *believe*, *assume*, *suggest*, *imply*, and *expect*; and hedging words, such as *likely* and *possibly*.

Examples:

- Teacher burnout *will likely continue* at a high rate.
- Early intervention *could mitigate* these effects.

Use the **present simple tense** to make recommendations or express opinions. If you are required to avoid using the first person, you may need the passive voice in this section.

Examples:

- We *suggest* repeating this study with a larger sample size.
- This study *indicates* that women who are at increased risk of pre-eclampsia would be willing to take steps to reduce that risk.

Use the **present perfect tense** to discuss past actions that are still relevant.

Example: Past legislation *has reduced* crime rates in similar circumstances.

3.3.7 Sentence length

Sentences that are too short and poorly connected can be irritating to read. Conversely, sentences that are too long and rambling are difficult to follow and are likely to be confusing. Use a sentence length that allows your thoughts to flow clearly. As a general rule there should be no more than 20-25 words in any one sentence. You may be able to reduce your sentence length by:

- cutting out unnecessary words

like might replace *along the lines of*

now may be just as appropriate as *at the present time*

we can now turn our attention to could perhaps be cut out entirely;

- dividing complex sentences into separate phrases or sentences.

If a breakdown occurs it is important that alternative supplies are available and the way that this is done is for the power stations to be linked through the high voltage transmission lines so that all of them contribute to the total supply of energy and an unexpectedly large demand can be handled.

can be re-written thus:

If a breakdown occurs it is important that alternative supplies are available; this is done by linking power stations through the high voltage transmission lines. All of them thus contribute to the total supply of energy and an unexpectedly large demand can be handled.

3.4 Citation Method

Quotations and foreign ideas should be highlighted within the theses/ARPs. Thereby, a uniform reference style is required throughout the whole thesis/ARP. Accordingly, quotations and foreign ideas must be highlighted as shown in the following:

- The references entail the name of the author and the year.

Example: „Hoeneß, U.: The advantages of the fiscal system in Switzerland, Munich 2007.“ appears as (Hoeneß 2007, p. 2).

- Direct quotations require quotation marks at beginning and the end of the reference. Implicit quotations (foreign ideas in own words) should be marked with „see“ in front of the reference.

Example „Möller, A.: Geography of south Europe – Insights from an expert, Dortmund 1992.“:

Direct quotation: „Milan or Madrid, the main point is that they are both in Italy.“ (Möller 1992, p. 123).

Implicit quotations: Shortly before the switch to Turin, the soccer player confused his fans with a statement which reveals his slight geographical knowledge (see Möller 1992, p. 123).

- To distinguish between several identical references, letters must be added behind the year: (see Hoeneß 2007a or see Hoeneß 2007b).
- If there are two authors (or editors), both are mentioned and separated by “&” or “/”.
Example: „Allofs, K.; Schaaf, T.: The influence of paper balls on matches – A case study. In: Applied soccer theories 22 (2009) 2, p. 357 - 370. appears as (see Allofs/Schaaf 2009, p. 42).
- If there are more than two authors (or editors), only the first author will be mentioned with the addition “et al.”.

Example: „Magath, F.; Meier, H.; Geyer, E.; Neururer, P.: Exhausting training methods for soccer player – Examples, Gelsenkirchen 2009.“ appears as (see Magath et al. 2009, p. 42).

- When citing a page (or a sentence/section) the exact page number is necessary: p. 215. Two pages: p. 215-216. Several pages: p. 317-325.
- If a section (more than one sentence) from one author will be cited, it must be marked with an opening sentence (Please consider that this reference applies only to the end of the section).

Example: In the following, the four communication types according to LEE ET AL. (2004, p. 35-39) are distinguished. First, continuously communication... .

- When naming authors in the sentence, the author name must be formatted as "small caps".

Example: MAGATH ET AL. (2009, p. 666) proposes medicine balls as tools to train soccer player.

- Missing detail about author, place or year should be assign to "a.u." (author unknown), "n.p.p." (no published place) or "n.y./n.d." (no year/no date).

3.4.1 References in the text

- Quotation of a section from the following book: „Meier, K.: Citation for experts, Munich 1911.“ → (see Meier 1911, p. 210-213).
- In addition, the following book is cited: „Meier, H.: Me and citations, Göttingen 1911.“ → (see Meier 1911a, p. 210-213) and (see Meier 1911b, p. 20-24).
- If there are several authors: „Wontorra, J.; Lattek, U.: Introduction into phrases, Freising 2009.“ → (see Wontorra/Lattek 2009, p. 43).
- If a sentence should refer to several books: „Beckenbauer, F.: Me as a ball, Munich 1972.“ and „Klinsmann, J.: My friend Lothar, Berlin 1997.“ → (see Beckenbauer 1972, p. 4; Klinsmann 1997, p. 109)³
- Quotation of a paper by Allofs and Schaaf in a journal → (see Allofs/Schaaf 2009, p. 357-365).

³ The order of the references should be oriented at the relevance of them.

3.4.2 Online references in the text

Online sources should be also highlighted. Thereby, the publication date of the source is used as the year.

- Example: The paper „Great soccer players like me“ from Franz Beckenbauer in the New York Times, published 1998. → (see Beckenbauer 1998)

If there is no author, you should select a suitable name.

- Example: A paper from the New York Times without an author published 1998. → (see New York Times 1998)

3.4.3 List of references

The references contain the following information:

- Books with one author:
name, first name: title, edition, location year. → (Meier 1911): Meier, K.: Citation for experts, Munich 1911.
- Books with several authors:
name author 1, first name author 1; name author 2, first name author 2; ... : title, edition, location year. → (Wontorra/Lattek2009): Wontorra, J.; Lattek, U.: Introduction into phrases, Freising 2009.
- Citation of a paper from Müller within a publication from Beckenbauer:
name, first name: title. In: name, first name (eds.): title, location year, pages of the paper. → (Müller 1975): Müller, G.: Aspects of penalty shoot-outs. In: Beckenbauer, F. (eds.): We are one, 29. edition, Munich 1975, p. 24 - 36.
- Quotation of a paper from Allofs and Schaaf, which was published in a journal:
name, first name: title. In: journal volume (year) issue, pages of the paper. → (Allofs/Schaaf 2009): Allofs, K.; Schaaf, T.: The influence of paper balls on matches – A case study. In: Applied soccer theories 22 (2009) 2, p. 357 - 370.
- Quotation of an online source:
name, first name: title. URL, publication date, request date. additional hints (if necessary). → The paper „Great soccer players like me“ from Franz Beckenbauer in the New York Times, published 1998. The request date was 1998-12-24 and the publication date

is 1998-07-01. → (Beckenbauer 1998): Beckenbauer, F.: Great soccer players like me. http://www.newyorktimes.com/Great_soccer_players_like_me.pdf, 1998-07-01, requested 1998-12-24. (Access only for subscribers).

The reference section is arranged alphabetically.

Examples:

- [Aga07] AGATHOS, A. et al.: *3D Mesh Segmentation Methodologies for CAD Applications*. In: Computer-Aided Design and Applications, Bd. 4 (2007), Nr. 6, S. 827-841
- [AAD98] ALEXANDER, P.; ALLEN, S.; DUTTA, D.: *Part Orientation and Build Cost Determination in Layered Manufacturing*. In: Computer-Aided Design, Bd. 30 (1998), S. 343-356

For further details regarding citation style and referencing, check the supplemental study material which is offered on the Moodle pages from Prof. Danjou.

3.5 Acknowledging, Citing and Referencing Use of AI Tools and Technologies

If your supervisor has advised that you can use generative AI tools and technologies in your thesis, this resource guides you on how to appropriately acknowledge the use of these tools and technologies in your assessments.

If you have used an AI tool or technology such as ChatGPT to generate an output which you either paraphrase or direct quote in your writing, you must cite and reference this output as a source in your reference list.

If you have used an AI tool or technology **in any other way in the process of completing your assessment** (for example brainstorming, understanding concepts, generating examples, summarising readings, seeking feedback on your writing), an acknowledgement of how you have used AI tools or technologies is mandatory. You can create this acknowledgement by adding a declaration at the end of your reference list.

A declaration is a personal statement of acknowledgement that you write to describe for readers how you have used AI tools and technologies to prepare an assessment. A declaration aims to create clarity and transparency about:

- the specific AI tools or technologies you used,
- what you used the AI tools or technologies for in the process of completing your assessment,

- the prompts you used in the AI tools or technologies,
- an explanation of how the output from the AI tools or technologies was used in your work.

For your readers, declarations add clarity about the extent to which you have used AI tools and technologies in your research, writing, and learning processes.

As a student, your declaration is an important statement confirming that the work you are submitting for assessment is your own and has academic integrity. Attempting to present AI-generated output as your original work without permission or proper acknowledgement puts you at risk of failure of the subject, suspension or expulsion.

The declaration generally follows the reference list and does not need to be started on a new page. It should be formatted to be consistent with other sections of your assessment in terms of font type, size and spacing. A heading ('Declaration') should be used to identify this statement clearly for your readers.

Here is a template for such declaration:

Declaration

I acknowledge the use of [AI tool or technology name] and [link] to generate.../ I have not used any AI tools or technologies to prepare this assessment.

Prompt: I entered the following prompt/s...

Use: I used the output to.../I modified the output to...

3.6 Further Comments in the Text

Footnotes should entail aspects that are not mandatory for the comprehensibility of the text. For example:

- Definition of terms, contextual delimitation and definition: „... the security ⁴of information systems...“.

⁴ In this case, the term “security” focuses on data privacy issues.

- Short examples and statistics: „...the most popular lectures ⁵in business information systems...“.
- Further explanation, mathematical evidence: „...the equation can be solved with the p-q-formula⁶...“.

3.7 Figures and Tables

Figures and tables help to visualize information. They should be integrated and mentioned in the text. In the case a table continues on the next page, the heading of the columns must be repeated. Furthermore, the text should refer to figures and tables, for instance (see Table 3-1), and should explain the illustrated content. Without such explanation, figures and tables are useless.

Table 3-1: Example of a caption for a table

Data	

The inscription of the tables and figures should be informative and be labeled like the following (below the figure and above a table). An inscription also belongs to tables and figures in the appendix.



Figure 3-1: Example of a caption for a figure

⁵ A survey among 2000 students reveals that the most popular lecture in business information systems is “Model and System development”.

⁶ The p-q-formula is a general method for the determination of zeros of polynomials of second order.

- Figure sequential number: Title of the figure (e. g.: Figure 1: Process of the seminar)
- Table sequential number: Title of the table (e. g.: Table 1: Differences in editing final papers)

When symbols or abbreviations are used, which require an explanation, figures and tables should entail a caption. Furthermore it is allowed to use a smaller font size as long it is readable. On top of that it is not necessary to use the text direction “justified”.

Additions, like “own illustration” at figures or tables are redundant. If a figure or table has no references, it is obvious, that it is an own illustration.

References of figures/tables generally don’t belong in the inscription, because they also do not belong in the List of figures/List of tables. They should be mentioned as a footnote or in a small font size under the figure or table.

3.8 Mathematical Equations

It is almost impossible to write a thesis in a scientific or engineering discipline without mathematical expressions. These expressions may contain mathematical functions, symbols, parameters, numbers, units or any combination of them. Here are some brief guidelines on how to write these mathematical expressions.

3.8.1 Formatting Mathematical Equations

A mathematical expression usually contains several parameters. All the parameters in the thesis have to be defined on their first appearance. However, they shouldn’t be the first word of any sentence. The following tips on formatting equations can help you in writing your mathematical expressions.

- Mathematical parameters should be written in italics, (not only in the equation itself but everywhere in the thesis), but not units, numbers and mathematical functions like logarithms.
- Font size for mathematical expressions can differ from the font size of the text. However, if possible (maybe there are limitations due to equation editors) use the same.
- The equations that are written on a separate line are either indented or centred, but done consistently throughout the thesis.

- Some equations are long and tend to be multi-line equations. If an equation breaks to multiple lines, align them properly and use only one number for the whole equation as far to the right as possible.
- Provide some vertical space above and below the equations that are displayed in separate lines.
- Equations are regarded as being a part of the text, and are punctuated accordingly. If it is the end of a sentence, there should be a period at the end of the equation; otherwise put a comma at the end of it.
- Do not capitalize or indent “where” or “with” when listing variables following an equation.

3.8.2 Numbering

There are two types of equations in a thesis: those which are going to be referred and those which are not. The first type of equation should be numbered consecutively, e.g. (3), or chapter-based, e.g. (3-1) through the whole thesis, whereas the second type can be either included in the running text if they are short and don’t break the flow of the sentence, or in a separate line without an equation number, if they are long and more complex. For numbering the first type of equations, i.e., those which are going to be referred somewhere in the thesis, here are some useful tips:

- The equation numbers should be enclosed by parentheses and placed at the right-hand side of the page.

Examples with chapterbased numbering:

$$(x + a)^n = \sum_{k=0}^n \binom{n}{k} x^k a^{n-k} \quad (3-1)$$

$$A = \pi r^2 \quad (3-2)$$

- If you are referring to an equation in the text, use eq. (3) (or other abbreviations). If it is the first word in a sentence, capitalize it, i.e., Eq. (3). Even better is to spell out “Equation” before the number, like “Equation (3)”.

Equations which are solved or reordered don’t get a new numbering.

3.8.3 Units

In a scientific or engineering thesis, it is very common to present the result of experiments, measurements or simulations. In all these cases, you need to provide numbers with their units. The following tips can help you to write them properly:

- The number and its unit should be followed on the same line.
- There should be a space between the number and the unit. There should also be a separator (space or /) between units when they appear together (as a product or division). However, there should be no space between the letters of units like millimeter “mm”.
- There shouldn’t be a space between a number and the percent, angular degree, angular minute, or angular second symbols (e.g. 35%, 90°, 15’, 4”).
- Use proper characters (sometimes special characters) for mathematical symbols.
- Units that accompany numbers should be abbreviated, like “10 m” but they should be spelled out when they do not accompany numbers, like “a few meters”.

Writing mathematical expressions without using an appropriate editor is difficult and sometimes impossible. Also, the result of not using it nearly always looks unprofessional. Microsoft Word comes with an equation editor which makes it easy to write almost any kind of equation. The other possibility is to use LaTeX which is a document preparation system developed specially for mathematical writing.

3.9 Guidelines Regarding the Format

The following format proposals should be used for a thesis and ARP:

- Margin: Top: 2.2 cm; left: 2.5 cm; right and bottom: 2 cm
- Font size: 12pt
- Line spacing: 1.2 (up to max. 1.3)
- Font: serif fonts such as Cambria or Times New Roman for the main text, sans-serif fonts such as Calibri for captions (title on cover page, chapter title, section title, ...)
- Text direction: Justified
- Headings should be brought out stylistically (bold, bigger font and so on)
 - Main chapters always start on a new page
 - Otherwise pages should be filled completely (no empty pages)

- Page numbers should be included in footers or headers.
- Footers:
 - Font size: 10pt
 - Line spacing: 1.2
 - Text direction: Justified
- Strong recommendation: use this document as a template! Replace all **red marked text passages** on the cover page and the *Declaration of Authorship* with your personal details and turn the color in black.

4 Common Mistakes in Theses and ARPs

4.1 Formal Mistakes

In the following, mistakes are shown, which are commonly found in theses and ARPs:

- Asymmetric structure as shown in Figure 4-1.

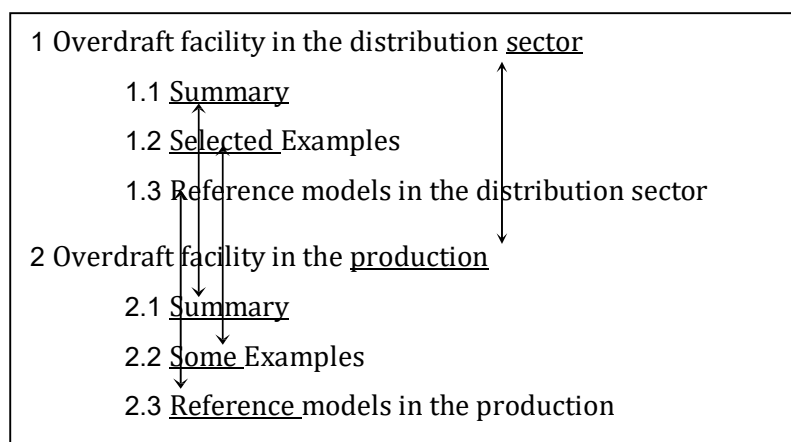


Figure 4-1: Example of an asymmetric structure

- Introduction is too long, and begins with general statements.
- References are inconsistent:

(Allofs/Schaaf 2009): Allofs, K.; Schaaf, T.: The influence of paper balls on matches – A case study. In: Applied soccer theories 22 (2009) 2, p. 357 - 370.

Müller, Gerd (1975): Aspects of penalty shoot-outs. In: Beckenbauer, F. (eds.): We are one, 29. edition, Munich 1975, p. 24 - 36.
- Sentences are too long and have a nested structure.
- Long sections without any structure.
- Inappropriate speech, e.g. colloquial or emotive language.
- Too many spelling mistakes and punctuation errors.
- Use of subjective language instead objective language.
- Using wrong tenses or an illogical mixture of tenses (scientific writing frequently uses the past tense).

4.2 Word Processing Hints

During the correction often seen and preventable formal mistakes:

- References with a break, where “p.” is on the current and the page number on the following page can be avoided by using a non-break space step (STRG+SHIFT+Space).
- Double spaces can be found and removed with the search function of IT-Tools.
- Don’t number figures, tables or equations manually. Use the appropriate Word functionality for automatic numbering.
- Inconsistent spaces between text and figures or tables.
- Inconsistent formatting such as mixture of font styles, bold, italic, etc.
- Spelling mistakes.

5 Evaluation

5.1 Evaluation Criteria for Theses

These are the criteria for evaluation of the final thesis report:

- Technical processing – Up to which extent did you reach the target?
- Use of technical knowledge
- Application of methods and tools
- Practicability – Can your results be implemented?
- Creativity
- Economic evaluation
- Self-dependence, initiative
- Systematic approach
- Documentation (report writing)
- Literature research
- Use of literature

5.2 Evaluation Criteria for ARP

These are the criteria for evaluation of the final ARP report:

- Technical processing – Up to which extent did you reach the target?
- Practicability – Can your results be implemented?
- Self-dependence, initiative
- Systematic approach
- Documentation (report writing)

5.3 Evaluation of the Colloquium

For the evaluation of the colloquium, the presentation as well as the question and answer session will be taken into account. Here are the evaluation criteria for the presentation part:

- Content of presentation
 - Structure (presentation in logical easy-to-follow format; definite opening, body and conclusion)
 - Relevance of material to topic set
 - Material presented indicated the speaker's knowledge and understanding of subject
 - Layout of slides (balance between text and figures)
 - Visual aids (legibility and suitability to enhance the presented work)
 - Presentation skills
 - Vocal quality: clear and articulated free speech; volume and speed of delivery; vocal variety
 - Presenting posture: gestures, body movement, facial expressions and eye contact
 - Speech reflects adequate preparation; allotted time has been effectively used
- List of references